

Groups, Orders, and Low-dimensional Dynamics

Summary table of persons involved in the project

Name	First name	Current position	Role & responsibilities	Involvement (person.month)
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Partner 1: Institut de Mathématiques de Marseille - UMR CNRS 7373				
TRIENTINO	Michele	PR (Marseille & IUF)	Coordinator, RT1, RT2	30p.month
DE CARPENTIER	Guillaume	PhD student (Marseille)	RT3	36p.month
GILBERT VIO	Martín	Post-doc (Rennes)	RT1, RT2	36p.month
LIOSSE	Isabelle	MCF (Lille)	RT2	12p.month
MATTE BON	Nicolás	CR (CNRS, Lyon 1)	RT1, RT2	24p.month
PUPULIN	Sandro	PhD student (Marseille)	RT1	24p.month
VIROT	Théo	PhD student (Dijon)	RT1	12p.month
WOLFF	Maxime	PR (Toulouse)	RT1, RT2, RT3	15p.month

Partner 2: Institut de Mathématiques de Jussieu-Paris Rive Gauche - UMR CNRS 7586				
GUIHÉNEUF	Pierre-Antoine	MCF (Sorbonne & ENS)	Partner's scientific leader, RT3	23p.month
ALVES	Pedro	PhD student (Sorbonne)	RT3	24p.month
EYNARD-BONTEMPS	Hélène	PR (Sorbonne)	RT1, RT2	23p.month
LE ROUX	Frédéric	PR (Sorbonne)	RT3	15p.month
MALICET	Dominique	MCF (Marne-la-Vallée)	RT2	3p.month
MILITON	Emmanuel	MCF (Nice)	RT2, RT3	30p.month

Recall that professors (PR) and associate professors (MCF) involvement only takes into account their research time, not their teaching time. Hence the maximal involvement for them is 30p.month.

The members of the projects will highly interact with Kathryn MANN and Andrés NAVAS, two of the leading figures of the field: Mann is now CNRS researcher (DR) at Sorbonne Université, while Navas will spend 3 years in Rennes thanks to the ANR *Choose France for Science* program. They are not formally in this project because they are involved in other research projects: Mann has submitted an ERC proposal, and Navas will be already funded by ANR.

Any changes that have been made in the full proposal compared to the pre-proposal

We have clarified the administrative structure of the project, keeping only two partners: in the pre-proposal we declared a partner for each institution to which a permanent member is affiliated. This was criticized by the committee: we certainly misunderstood a recommendation from the administration of Aix-Marseille Université. Contrary to what declared in the proposal, Partner 2 is now asking funding: this is a more natural choice, considering that several people are affiliated to Sorbonne University. After this modification, we find that it is more strategic to have one of the two post-docs in Paris. However, the cost of the salary in Paris is higher compared to the one in Marseille, and so we have to reduce the offered post-doc to 1 year. This explains the 8% reduction of the budget, which is a bit more than the allowed 7%.

I. Proposal's context, positioning and objectives

I.A. Objectives and research hypothesis

The aim of this project is to make fundamental contributions to the very active subject of **group actions on low-dimensional manifolds**, through the point of view of dynamical systems. This is one of the main aspects of the emerging field of **dynamical group theory**, whose purpose is to *study groups through actions on manifolds*, in the same way as classical geometric group theory aims at studying groups through actions on metric spaces. By its nature, the study is related to several fields, such as *group theory*, *geometric topology*, and *dynamical systems*, strongly interacting together. When studying the dynamics of groups of transformations of a space, the space itself need not be complicated to observe rich phenomena: we will consider group actions in dimension 1 and 2 (considering higher-dimensional actions as a whole theory is currently out of the reach).

The theory in dimension one is well established, with roots in the theory of *orderable groups* and *foliation theory*; several monographs have been written in the last 25 years [19, 24, 29, 39, 59, 60, 78]. Recent years have been prolific, in part thanks to the impulsion generated by Navas' 2018 ICM address [80]; this activity has led to a change of paradigm. Notable advances include the solution of the longstanding problem of non-orderability of high-rank lattices [26] and the surprising solution to the Wiegold problem [22]. New tools allow us to address finer questions on the description and classification of actions [17, 28, 34]. The importance of the action's regularity is better understood [58, 73], and several examples have revealed new phenomena [15, 56, 75]. In contrast, the study of two-dimensional actions has started setting solid bases only in recent years, thanks to the *breakthrough work* of Bowden-Hensel-Webb [10]. The research proposal is structured with three research tracks (RTs). Here is a rough summary.

- RT1** A current trend in the study of one-dimensional actions is to study **representation spaces** defined by considering all possible actions of a given finitely generated group. We aim at describing such spaces for relevant examples. We also want to use them as a *tool* to study how orderability is related to group properties such as amenability, Haagerup property or Kazhdan's property (T), and questions around orderability of 3-manifold groups (motivated by the famous *L-space conjecture*).
- RT2** We propose to study **one-dimensional actions in a broader sense**: groups of one-dimensional germs and groups of homeomorphisms of non-Hausdorff one-manifolds. So far there have been only scattered results, and we aim at a more structured theory. One motivation for this is that actions of this nature are appearing *more and more frequently*, especially through constructions in geometric topology (such as leaf spaces of codimension-one foliations of 3-manifolds), but not only.
- RT3** In the context of dimension two, there is **fundamental work** to do around the *fine curve graph* introduced by Bowden, Hensel, and Webb. It is a Gromov-hyperbolic space on which the group of homeomorphisms of the surface acts, in a non-elementary way. Considering the intense activity on this subject, it is highly expected that in a few years it will be a *very powerful tool* to get structure results on group actions on surfaces.

When describing the different tasks, we give an estimation of the difficulty using a certain number of asterisks: (*) means that this requires some work, but we know how to do it; (**) indicates that the problem is difficult, but we have some clues on how to solve it; (***) indicates a problem which is difficult to solve, and we do not have very precise clues on how to attack it.

I.A.1. RT1: Spaces of one-dimensional actions

Objective 1: Amenability, Haagerup, and Kazhdan properties Many fundamental problems for groups acting on 1-manifolds have been solved in recent years, and somehow our approach to the subject has deeply changed. However, there are still very basic important questions which remain unsolved. Here we will focus on the connection between one-dimensional dynamics and analytic properties of groups, mainly motivated by the following well-known, highly challenging problems:

Problem 1. *Is the group $PL(I)$ of piecewise-linear homeomorphisms of the interval $I = (0, 1)$ amenable?*

Problem 2. *Are there infinite, finitely generated subgroups of $\text{Homeo}(S^1)$ with Kazhdan's property (T)?*

These problems have a long history. Problem 1 is a bit stronger than the famous problem of understanding whether *Thompson's group* F (a subgroup of $\text{PL}(I)$ formed by elements defined by dyadic rational parameters) is amenable. Recall that a group is **amenable** if every action by homeomorphisms on a compact space admits an invariant probability measure. Groups of sub-exponential growth are amenable, and the class of amenable groups is invariant under the operations of taking extensions, subgroups, and direct limits. Groups that can be obtained with these operations starting from abelian groups are called **elementary amenable**. The standard example of non-amenable group is the non-abelian free group F_2 , and it took several decades to find examples of non-amenable groups without copies of F_2 . By a result of Brin and Squier (see [39, Thm 4.6]), the group $\text{PL}(I)$ does not contain any F_2 . Quite surprisingly, Monod has proved that the group $\text{PP}(\mathbb{R})$ of *piecewise-projective* homeomorphisms of the line is non-amenable and contains no F_2 (see [29, Cor 4.2.5]). On the other extreme we find **Kazhdan's property (T)**; for countable groups, it can be seen as a fixed point property for isometric affine actions on Hilbert spaces. This is often considered as a *rigidity property* in group theory, and the class of groups with (T) is much more restricted. The standard examples are irreducible lattices in higher-rank semi-simple Lie groups, such as $\text{SL}(n, \mathbb{Z})$ for $n \geq 3$. Navas has proved that the group $\text{Diff}^{3/2}(S^1)$ of circle diffeomorphisms of class $3/2$ has no infinite, finitely generated subgroup with (T) (see [78, Thm 5.2.14]). The question is wide open for groups of diffeomorphisms in lower regularity, although recently, Deroin and Hurtado have proved that higher-rank lattices as above do not embed in $\text{Homeo}(S^1)$ [26]. We will discuss some of the approaches to these problems inspired by tools introduced recently for one-dimensional actions. But before this, we want to mention a related third problem, that we present as our first challenge within Objective 1:

Task 1 ().** *Is there a bi-orderable group without the Haagerup property?*

More concretely, we can ask:

Task 2 ().** *Which subgroups of $\text{PL}(I)$ and $\text{PP}(\mathbb{R})$ have the Haagerup property? Which admit a proper action on a $\text{CAT}(0)$ cube complex?*

We have learnt the question in Task 1 from Navas, whereas the related Task 2 was proposed by Dinamarca, López Neumann, and VIROT and they are currently working on it. Recall that a group is **bi-orderable** if it admits a total order which is invariant under left and right multiplication. The group $\text{PL}(I)$ is bi-orderable. A weaker notion is **left-orderability**, for which invariance is required only under left-multiplication. These properties are clearly preserved when considering subgroups. The group $\text{Homeo}_+(\mathbb{R})$ of orientation-preserving homeomorphisms of the line, and thus all of its subgroups, are left-orderable. Conversely, any *countable* left-orderable group embeds in $\text{Homeo}_+(\mathbb{R})$. A (countable) group has the **Haagerup property** if it admits a *proper* isometric affine action on a Hilbert space. Among groups with the Haagerup property, we find amenable groups, but also free groups, and more generally any group admitting a proper action on a $\text{CAT}(0)$ cube complex (notably, Thompson's group F ! [54]). Clearly, a group with the Haagerup property and (T) is finite. Usually the lack of Haagerup property is witnessed by some sort of relative property (T), and in fact there are only few known examples of left-orderable groups without (T) and the Haagerup property at the same time (and the reason is that we don't know much about Problem 2). A similar problem was proposed by Navas in the context of groups of circle diffeomorphisms [76], inspired by his work on property (T) mentioned before, but we believe that the PL setting is more treatable: in the last years many authors have produced actions of groups of PL homeomorphisms on $\text{CAT}(0)$ cube complexes and thus isometric affine actions on Hilbert spaces, whereas in the differentiable setting there has been no advance since the work of Navas.

For the next tasks, we have to introduce an object that has completely changed the approach to the study of finitely generated group actions on the line: the structure, and the restrictions for actions on the line of a given group are studied by considering the properties that the *space of all possible actions* should have. Formally, one considers the *space of normalized harmonic actions*, introduced by Deroin [25] and studied extensively in [17], also called **Deroin space** in [15]. Let us define it. By an action of a group G on the line, we mean a homomorphism $\varphi: G \rightarrow \text{Homeo}(\mathbb{R})$. With almost no loss of generality we can simply consider *orientation-preserving* actions (so homomorphisms into $\text{Homeo}_+(M)$). We say that an action is *irreducible* if there is no global finite orbit. From the point of view of dynamical systems, two actions are basically the same if one is obtained from the other after a change of variable (that is, by a *conjugacy*), but we want also to identify two actions if one is obtained from the other by collapsing and blowing up orbits (that is, by a

semi-conjugacy). Formally speaking two actions $\varphi, \psi: G \rightarrow \text{Homeo}_+(\mathbb{R})$ are (positively) semi-conjugate if there exists a proper non-decreasing map $h: \mathbb{R} \rightarrow \mathbb{R}$ such that $h\varphi(g) = \psi(g)h$ for every element g of G . We denote by $\text{Rep}(G, \mathbb{R})$ the space of irreducible actions of G on $\mathbb{R}$ up to semi-conjugacy, which can be seen as the “one-dimensional non-linear representation theory” of G . The Deroin space $\mathcal{D}$ of G can be abstractly defined as the space of irreducible actions of G on $\mathbb{R}$ up to *pointed* semi-conjugacy, which is refinement of semi-conjugacy defined by requiring that the non-decreasing map h takes the marked point 0 to 0. Given any two irreducible semi-conjugate actions φ and ψ , there exists a unique translation $T_t(x) = x + t$ such that φ is pointed-semi-conjugate to the action $T_{-t}\psi T_t$; consequently, the space $\text{Rep}(G, \mathbb{R})$ corresponds to the orbit space of the **translation flow** $\Psi: \mathbb{R} \times \mathcal{D} \rightarrow \mathcal{D}$, which is defined as $\Psi^t(\psi) = T_{-t}\psi T_t$.

Working with the Deroin space $\mathcal{D}$ instead of $\text{Rep}(G, \mathbb{R})$ has several advantages. Both spaces are compact, but when considering the topology on these spaces induced by the compact-open topology on the subset of irreducible representations in $\text{Hom}(G, \text{Homeo}_+(\mathbb{R}))$, we have that $\mathcal{D}$ is metrizable, whereas the topology of $\text{Rep}(G, \mathbb{R})$ can be pretty bad (for instance, it can be homeomorphic to the orbit space of a subshift [15]). In fact, after the fundamental work [28] on harmonic measures for symmetric random walks on $\text{Homeo}_+(\mathbb{R})$, one has that the Deroin space $\mathcal{D}$ is homeomorphic to a *compact subspace* of irreducible representations, invariant under the translation flow (this is the space of normalized harmonic actions). Moreover, dynamical properties of the translation flow allow us to understand how semi-conjugacy classes behave under perturbations [15, 17]. One also gets a clear understanding of the nature of actions: fixed points of Ψ correspond to non-trivial homomorphisms $G \rightarrow \mathbb{R}$, or equivalently to actions semi-conjugate to translations; closed Ψ -orbits correspond to irreducible actions with non-trivial centralizer, that is, actions that are the lift of actions on the circle. When a Ψ -orbit is not closed, its ω -limit describes how an action on this orbit behaves near $+\infty$.

The original motivation of Deroin for considering this space of actions was to give a more conceptual proof of the result of Witte Morris:

Theorem 1. *Any finitely generated (non-trivial) amenable subgroup of $\text{Homeo}_+(\mathbb{R})$ surjects to $\mathbb{Z}$.*

In fact, that theorem can be restated as follows: for such a G , the closure of every orbit of the flow $(\mathcal{D}, \Psi)$ contains a fixed point. However, in all known examples of amenable groups acting on the line, a stronger property holds: every Ψ -orbit *converges to a limit* (necessarily a fixed point of the flow). We do not know if this is true for all amenable groups, and other similar questions can be asked. The following goal resumes in a nutshell a family of projects in this direction, currently investigated by Brum, GILBERT VIO, MATTE BON, Rivas, and TRIESTINO.

Task 3. *Study restrictions of the translation flow Ψ on the Deroin space $\mathcal{D}$ for finitely generated groups in the following classes:*

- *amenable groups (***)*;
- *elementary amenable groups (**)*;
- *groups with the Liouville property for the simple random walk (**)*.

Sample questions: for each of these classes, do all Ψ -orbits converge to a limit? Does the flow Ψ admit recurrent points (that are not fixed)?

For general amenable groups, we are inclined to believe that the question has a negative answer. In fact, we can prove that a positive answer would imply the non-amenability of F (although F itself satisfies this property [17]), and thus solve Problem 1. Finding a counterexample seems to require new ideas to construct amenable groups acting on the line. However, we believe that the answer might be positive for elementary amenable groups (evidence is that it is true for solvable groups [16]), and the framework of *laminar actions* from [17] (we will discuss more about this notion later) provides a possible approach. It is also plausible that the *Liouville property* (every bounded harmonic function is constant) implies additional restrictions on the space $(\mathcal{D}, \Psi)$, since both are related to harmonic measures.

A very concrete problem that may lead to new constructions of amenable groups is the following:

Task 4 (*)**. *Are there examples of finitely generated amenable groups admitting an action on the line which is not laminar and not semi-conjugate to any affine action?*

Note that the standard action of Thompson’s group F is of this nature, but clearly we don’t know if it is amenable!

The Deroin space is a useful tool even for groups that do not act on the line! Deroin and Hurtado have used its abstract properties in their proof that lattices in higher-rank simple Lie groups do not act on the line [26]: before this work, it was known from results of Burger–Monod and Ghys (see [39, §7.2]) that the translation flow has no fixed point and no closed orbit. In fact, no example of finitely generated group with this property was known until very recently (see [80, Question 4]): the first examples have been produced by MATTE BON and TRIESTINO (reworking the construction of [56]). The main idea is a *change of perspective*: given a compact metrizable space X with a topological flow Φ defined on it, one can consider groups of homeomorphisms of X that preserve single Φ -orbits. We will try to use this different perspective to get new insights into Problem 2, trying to construct left-orderable groups with property (T), or conversely to find more constraints for such groups.

A relatively easier task concerns *elementary amenable subgroups of PL homeomorphisms*. It is known that Thompson’s group F is not elementary amenable, but it is believed that it is minimal with respect to this property.

Task 5 ().** *Prove that if a finitely generated subgroup of $PL(I)$ contains no subgroup isomorphic to F , then it is elementary amenable. Conversely, prove that any finitely generated elementary amenable subgroup of $PL(I)$ is conjugate to a subgroup of Thompson’s group F .*

The first part has been conjectured by Brin and Sapir [14], while the second part does not appear explicitly stated in the literature, but there are many supporting evidences. Our approach is to combine the structure results on groups of PL homeomorphisms from the works of Bleak–Brin–Moore [7] and Bleak–Brin–Kassabov–Moore–Zaremsky [6] with the theory of laminar actions introduced in [17] and applied in [16] to solvable groups. Let us say that an irreducible action on the line is *laminar* if it is semi-conjugate to an action that preserves a **lamination**: this is a family of bounded open intervals which are pairwise either nested or disjoint (visually, the corresponding half-circles drawn on the upper half-planes are pairwise disjoint). Invariant laminations for actions on the circle are a classical objects, inspired by constructions in hyperbolic geometry (see for instance [19]); when considering laminar actions on the line, in several situations the invariant lamination allows one to have a simpler description of the action outside a sufficiently large bounded interval. The new insight is an application of a result from [17]: if a non-solvable finitely generated subgroup of $PL(I)$ acting minimally does not contain any copy of F , then its action on $I \simeq \mathbb{R}$ is laminar. Note that the situation for solvable groups is completely understood, thanks to work of Bleak (which can be probably recovered using the theory of laminar actions).

The second part of Task 5 seems easier: a result of [6] says that if a finitely generated subgroup of $PL(I)$ admits a *geometrically fast* generating set, then it is conjugate to a subgroup of F . Thus, it remains to prove that if a subgroup does not admit any geometrically fast generating set, then its action cannot be laminar. There are however several technical problems: the main one is that the action of subgroup on I may not be minimal, and then one has to do some *level theory*. The same problem occurs when we want to follow a strategy for solving the first part of Task 5: if a subgroup G of $PL(I)$ has a laminar action on I , then there exists an element $f \in G$ and a sub-interval $I' \subset I$ such that $f^n(I')$ tends to I as $n \rightarrow +\infty$ and $f^n(I')$ tends to some compact interval $K \subset I$ when $n \rightarrow -\infty$. We say that f is an (expanding) *pseudo-homothety*, and when K is a single point we say that f is a homothety. When G contains homotheties, we want to argue that G is conjugate to a subgroup of the so called Brin–Navas group, which is the minimal non-solvable finitely generated subgroup of $PL(I)$. Otherwise, we want to define some complexity for the pseudo-homothety, using the compact interval K and the dynamics of compactly supported elements of G , in such a way that it is possible to describe the action of G in terms of an action of another subgroup of $PL(I)$ of lower complexity, by looking at the action on the invariant lamination (and in such a way that this operation stops after finitely many steps).

Objective 2: Orderability and 3-manifolds groups The study of orderability properties of fundamental groups of 3-manifolds is a classical topic, and we refer to the book of Clay and Rolfsen [24] for an introduction. One of the first notable results was by Boyer, Rolfsen, and Wiest (see [24, Thm 5.1]):

Theorem 2. *Suppose M is an orientable, irreducible 3-manifold, with $M \neq S^3$. Then $\pi_1(M)$ is left-orderable if and only if it admits a non-trivial homomorphism into $\text{Homeo}_+(\mathbb{R})$.*

In particular, any such manifold with positive first Betti number has left-orderable fundamental group. This basically gives that *virtually* (that is, up to pass to a finite index subgroup), every such fundamental group is left-orderable. From the perspective of geometric group theory, this would be a satisfactory statement, but it turns out that the problem of determining *exactly* when a 3-manifold group is left-orderable has been producing considerable work for the last 15 years. The reason is a surprising prediction of Boyer-Gordon-Watson and Juhász (see [13]), known as the ***L-space conjecture***:

Conjecture 1. *Suppose M is a closed orientable, irreducible 3-manifold, with $M \neq S^3$. Then the following are equivalent:*

1. $\pi_1(M)$ is left-orderable,
2. M carries a co-orientable taut foliation,
3. M is not a *L-space*.

Let us quickly introduce the objects in this statement: a **taut foliation** of a 3-manifold is a codimension-one foliation admitting a transversal circle intersecting any leaf. When the first Betti number of the 3-manifold is positive, the existence of a taut foliation is guaranteed by an old result of Gabai (see [19]). An ***L-space*** is a *rational homology sphere* (that is, with zero first Betti number), whose Heegaard-Floer homology is minimal. The Heegaard-Floer homology of a rational homology sphere is an analytic object, and its topological meaning is somehow obscure. The *L-space conjecture* would provide a more intuitive definition of an *L-space* in terms of properties of the fundamental group (not left-orderable) and topological properties (no co-orientable taut foliation). So far, the only established implication is $2 \Rightarrow 3$, proved by Ozsváth and Szabó [82], but the conjecture has been verified in several particular cases. One notable case is the class of **graph manifolds**, solved by Boyer-Clay [12] and Hanselman-Rasmussen-Rasmussen-Watson [51].

Although many people have been working on the *L-space conjecture*, most of them come from geometric topology, and have a little experience with the tools of general group actions on the line. So far, one of the most successful strategies for producing actions on the line is to get first an action on the circle from constructions in geometric topology (see for instance [18, 22, 69]), and then try to compute if the corresponding Euler class vanishes, thus permitting to lift the action to the line. In the terminology of the Derooin space introduced in Objective 1, this means that the investigation is often *reduced to closed orbits of the translation flow*. An interesting step away from this line of exploration appears in the work of Calegari and Loukidou [22], when they shortly discuss *uniform actions*.

Task 6 ().** *Use the Derooin space to get new insights into the *L-space conjecture*.*

In fact, Theorem 2 suggests that the setting of the Derooin space is certainly more appropriate for investigating the *L-space conjecture* than the more classical space of left-orders: the former takes into account *non-trivial minimal actions on the line*, up to conjugacy, whereas the latter takes into account all *faithful irreducible actions*.

The PhD student Sandro PUPULIN has started working this year under the supervision of Luisa Paoluzzi and Michele TRIESTINO, and the goal of his thesis is to describe the Derooin space for (left-orderable) fundamental groups of rational homology spheres, at least in cases where the *L-space conjecture* has been verified. Unfortunately, PUPULIN has started his PhD without a scholarship, and we ask a salary for the remaining 2 years in order to let him work in a more comfortable situation.

As a preliminary result, PUPULIN was able to verify, using classical results of Naimi and Jankins-Neumann that when the manifold is Seifert fibered, then the Derooin space is simply described by two periodic orbits. The work of Boyer-Clay [12] should allow him to get a description for the next natural class:

Task 7 (*). *Describe the Derooin space for fundamental groups of graph manifolds that are rational homology spheres.*

Another task that recalls the discussion on uniform actions of Calegari and Loukidou is to find examples of rational homology spheres whose associated translation flow on the Derooin space has non-periodic orbits. More generally, we want to consider the following problem:

Task 8 ().** *Determine for which rational homology spheres the translation flow on the Derooin space has only periodic orbits.*

Note that since fundamental groups of rational homology spheres have finite abelianization, the translation flow on the Deroin space has no fixed point. Therefore, a non-periodic orbit will have a non-trivial ω -limit, and this leads to actions on the line of such groups with interesting dynamical behavior.

Let us also mention here a problem about actions on the circle of *hyperbolic 3-manifold groups*. In his work on closed orbits for quasi-geodesic flows on hyperbolic 3-manifolds [36], Frankel observes that an alternative proof would follow if one is able to show that the fundamental group admits no hyperbolic-like action on the circle. Here we say that an action on the circle is **hyperbolic-like** if any non trivial element acts with exactly two fixed point, one repelling and one attracting. The question of Frankel has led Kim and TRIESTINO to study more generally groups of hyperbolic-like circle homeomorphisms [57]. Typical examples of hyperbolic-like groups are given by discrete subgroups of $\mathrm{PSL}(2, \mathbb{R})$, but there are also examples of different nature obtained by Kovačević by surgery [62].

Task 9 ().** *Obtain a structure theorem for hyperbolic-like groups of circle homeomorphisms. For instance, do they always preserve a lamination of the circle? Are all the non-Fuchsian examples obtained by variants of the Kovačević construction?*

A positive answer to this question would also allow to reprove Frankel's theorem on closed orbits of quasi-geodesic flows.

Objective 3: Spaces of differentiable one-dimensional actions The study of group actions on one-manifolds has a long tradition of considering *regularity issues*: what are the algebraic constraints that arise when considering actions by diffeomorphisms of class C^r , for a given $r \geq 0$? One of the first remarkable results is the famous Kopell's lemma (see [78, §4.1]), after which the centralizer of a C^2 diffeomorphism of the closed interval is isomorphic to a subgroup of $\mathbb{R}$. The book of Navas [78] has contributed to generate an intense development of the study of regularity in the last 15 years. Let us mention the recent surprising work of Kim and Koberda [58] (see also Mann and WOLFF [73]) exhibiting, for any regularity $r \geq 1$, a finitely generated subgroup of $\mathrm{Diff}_+^r(S^1)$ which is not isomorphic to any subgroup of $\mathrm{Diff}_+^s(S^1)$, for any $s > r$.

As for the case of actions by homeomorphisms, there is a growing interest in understanding the **topology of the space of differentiable actions** of a given finitely generated group on a one-manifold M . However, when comparing the case of differentiable actions with that of topological actions, we see that there is no way of reducing the study to representative of semi-conjugacy classes (like we did when introducing the Deroin space in Objective 1), since semi-conjugacy does not well behave with respect to regularity (as one can see for instance with the classical theorem of Denjoy, or with Kopell's lemma). The study of spaces of differentiable actions is way more complicated, and for this reasons the groups under consideration and the list of questions are much less sophisticated. We have for instance the following notoriously difficult problem:

Problem 3. *Is the space of C^∞ actions of $\mathbb{Z}^d$ ($d \geq 2$) on $[0, 1]$ path-connected?*

According to EYNARD-BONTEMPS's thesis [31], a positive answer would imply path-connectedness of the space of smooth codimension-1 foliations in a given homotopy class (of plane fields) on any 3-manifold, yielding a parametric version of a famous result of Thurston and hinting towards a full h-principle for foliations.

It should be clear from our initial discussion that the *topological properties of spaces of differentiable actions are highly sensible* to the group and the regularity parameter. In the C^1 setting for instance, it is known that the space of actions of abelian or even nilpotent groups is path-connected [79, 83]. The case of groups of intermediate growth represents a challenge. Recently, Kim-MATTE BON-de la Salle-TRIENTINO [61] proved that every continuous action on the interval is semi-conjugate to a C^1 action, and these actions can be produced so that the generators lie in any prescribed neighborhood of the identity. Whether or not the whole space of actions is path-connected (or even connected) is a challenging question.

Task 10 ().** *Show that the space of C^1 actions on a compact connected one-manifold of a group of intermediate growth is (path-)connected.*

This would require to fully understand the mechanisms that yield exponential growth for orbits and/or derivatives of group elements. Along this path, this should yield to a correct proof of a famous claim of Hurder that zero entropy is equivalent to having no resilient orbit for a C^1 action (see [55]; the case of C^2 actions

was solved by Ghys–Langevin–Walczak [40], but this is way simpler, because the level theory of C^2 actions is very rigid). We are not going to define the entropy of a one-dimensional action here (whose definition, as for any kind of entropy, is quite involved), but one may think of it as a way to measure the exponential growth of orbits. This statement is the analogue, for one-dimensional actions, of the well-known theorem of Katok that a surface diffeomorphism has positive entropy if and only if there is a horseshoe. Indeed, a resilient orbit is the correct analogue of a horseshoe, and it is easy to see that it ensures positive entropy for the action.

Task 11 ().** *Prove that if a C^1 action of a finitely generated group on the circle has no resilient orbit, then its entropy vanishes.*

Going back to spaces of differentiable actions, we have that when we increase regularity, the situation is even more unclear. In this context, on the one hand, there is less place to perform deformations, but on the other hand there are fewer actions. This search of equilibrium is exploited by EYNARD–BONTEMPS and Navas in [34], where they proved that the space of C^{1+ac} (resp. C^2) actions of abelian groups on connected compact 1-manifolds (resp. the interval) is path connected (resp. connected). A natural question is whether this extends in class C^2 . However, this is a challenging problem, which seems to be related to deep phenomena around this regularity class, as for example the famous problem of (non-)simplicity of the full group of C^2 diffeomorphisms.

I.A.2. RT2: One-dimensional actions, in a broader sense

Objective 4: Groups of germs In a related direction, **groups of germs of diffeomorphisms** deserve more attention. We will denote by $\mathcal{G}^r = \mathcal{G}^r(\mathbb{R}, 0)$ the group of (one-sided) germs of C^r diffeomorphisms of the line fixing 0. Recently, Navas noticed that groups of germs of homeomorphisms at the origin are left-orderable (see [29, Rmk 1.1.15]). This implies that every such group admits an action on the interval if finitely generated. However, these realizations are not explicit and, in concrete examples, most of them tend to destroy regularity. In the other way around, the natural map from the group of homeomorphisms of the unit interval to the group of germs at the origin is not injective. Thus, results that are valid for groups of diffeomorphisms are not ensured to hold for groups of germs, particularly whenever the proof involves the whole action on the interior. As an explicit situation to consider, we have Navas’s alternative for finitely generated groups of C^r diffeomorphisms ($r > 1$): such a group is either virtually nilpotent or contains a free subsemigroup [77].

Task 12 ().** *Let $G < \mathcal{G}^r$ be a finitely generated group of C^r differentiable germs without free subsemigroups. Is G virtually nilpotent?*

Such a result has potential applications in the context of codimension-one foliations, where groups of germs naturally arise as holonomies around closed leaves.

Another interesting problem involving germs concerns **distortion elements**, namely elements that fit inside finitely generated subgroups of the involved group in such a way that the word-length of their iterates grows sublinearly with respect to the exponent.

Task 13 ().** *Characterize distortion elements in $\mathcal{G}^r$. What about the group $\mathcal{G}^\omega$ of real-analytic germs?*

Distortion elements in groups of C^1 , C^2 and/or C^∞ diffeomorphisms of either the circle or the interval are well understood thanks to works of Avila, Navas [81], EYNARD–BONTEMPS–Navas [33], and EYNARD–BONTEMPS–MILTON [32]. Again, the case of germs is widely open, and this is also the situation in a particularly interesting case, namely that of the group of **germs of real-analytic diffeomorphisms**. Therein, the only distortion elements that are known are those that are conjugate to parabolic elements of the naturally embedded affine group. However, it is hard to imagine that these are the only ones. The heuristic comes from the case of C^2 diffeomorphisms of the closed interval, for which Navas introduced an obstruction to distortion, called *asymptotic variation*. As EYNARD–BONTEMPS and Navas proved later, this number coincides with a numerical feature of a widely known conjugacy invariant, the so-called *Mather invariant*. Now the point is that, for germs of real-analytic maps, there is an analog of this invariant, the *Écalles–Voronin invariant*. Given the available results in the interval case, it would be natural to conjecture that parabolic germs with non trivial Écalles–Voronin invariant are undistorted, yet this is perhaps a naive guess. In any case, getting some understanding on the problem is already challenging.

We conclude this discussion with some problems of different nature, which are somehow related to Objective 2. Recently, a team of people in Rennes has proved that surface groups embed in the group $\mathcal{G}^\omega$ of real-analytic germs [21]. A more general problem asked by Cerveau and Ghys is to determine whether a surface group embeds in the group $\text{Diff}_+^\omega([0, 1])$ of real-analytic diffeomorphisms of the interval. Trying to go in the opposite direction, we propose to prove the following:

Task 14 (*). *If Γ is the fundamental group of a closed hyperbolic 3-manifold, then it does not embed in $\mathcal{G}^\omega$.*

This would also represent a first step for solving a rigidity problem suggested by Sang-hyun Kim:

Task 15 ()**. *Let Γ be the fundamental group of a closed hyperbolic 3-manifold. If $\varphi: \Gamma \rightarrow \text{Diff}_+^\omega(S^1)$ is a faithful action, then φ is topologically conjugate to a representation of Γ in $\text{PSL}(2, \mathbb{R})$ (defined by a Galois automorphism of the trace field for its natural representation in $\text{PSL}(2, \mathbb{C})$).*

Objective 5: Actions on non-Hausdorff one-manifolds Considering group actions on non-Hausdorff one-manifolds may sound a bit pathological, but it is actually motivated by recent works on laminar actions (see the discussion in Objective 1) and familiar constructions in geometric topology, such as leaf spaces of codimension-one foliations, or the bifoliated planes appearing in the theory of pseudo-Anosov flows on 3-manifolds (see for instance [2]), which lead to actions of fundamental groups on leaf spaces, which are typically non-Hausdorff one-manifolds. For us, considering actions on such manifolds is also relevant in the perspective on what we discuss in RT3, see notably Task 23 and Objective 9: when a group action on a surface preserves some partition, there is an induced action on the corresponding leaf space which is a one-dimensional object, but typically not Hausdorff.

Task 16 ()**. *Develop a general theory of group actions on non-Hausdorff one manifolds.*

We will next describe more precise questions. The 2-year post-doc in Marseille will work on several aspects of this Objective, mainly in collaboration with TRIESTINO, but also sharing ideas with other members such as LIOUSSE, MALICET, MATTE BON, and MILITON.

A pioneering work in this setting was produced by Barbot more than 30 years ago [1], but the theory has not been really developed. Because of the lack of a general theory, several recent works in geometric topology often contain ad-hoc partial results (see for instance [20]). Note that, upon passing to the universal cover of the non-Hausdorff one-manifold, we can assume that the manifold is simply connected, and thus its Hausdorffication is somehow a **topological real tree** (by this, we mean that the underlying space carries a real-tree distance, but we consider actions by homeomorphisms on it). Although considering actions by homeomorphisms on real trees may sound less pathological, there may be a loss of information in the Hausdorffication process: for instance, in the recent work [85], Zhao considers the action of the fundamental group on the leaf space of the lift to the universal cover of a co-orientable taut foliation of a 3-manifold, and under some assumption he is able to use the non-Hausdorff nature to define a left-invariant order on the fundamental group.

A first natural question is inspired by a result in the work of Barbot [1], that extends to this setting the result of Witte Morris that finite index subgroups of $\text{SL}(n, \mathbb{Z})$, for $n \geq 3$, are not left orderable.

Task 17 ()**. *Prove that if Γ is a lattice in a higher-rank simple Lie group, then every action of Γ on a non-Hausdorff one-manifold has a global fixed point.*

This would represent an extension of the classical Serre's property FA for actions by homeomorphisms, and the question already appears in [1]. Our strategy is to apply the ideas of Deroin and Hurtado [26] to this setting. There may be an easier strategy: when such an action preserves an end, then there is a well-defined morphism of Γ into the group $\mathcal{G}_\infty$ of germs at infinity of the line (this has been remarked by Zhao [84], but it is also implicit in the works [16, 17] discussing laminar actions on the line). Since the group $\mathcal{G}_\infty$ is left-orderable (as we mentioned in Objective 4), the result of Deroin and Hurtado prevents this situation. If there is no fixed end, then the induced action on the space of ends should be a nice *boundary action*, and one can hope to use the sophisticated, but now classical, theory of Furstenberg.

A second natural question is inspired by an old work of Levitt [67], where he considers non-nesting actions on topological real trees. An action is *non-nesting* if whenever an arc is sent inside itself by an element of the group, then the element fixes the arc. Examples of non-nesting actions are given by *free actions*. Levitt

proves that if a finitely presented group has a non-nesting action on a topological real tree, then the action preserves a Radon measure. This is an analogue of the classical theorem of Hölder for free actions on the real line, but there is the annoying assumption of finite presentation, which comes from the fact that Levitt relies on the theory of *group actions on real trees*, developed by Rips and Sela. A key fact in this theory is that an isometric action on a real tree can be encoded by a group of *interval exchange transformations* (see for instance [37,38]). For more general actions on topological real trees, this machinery would produce a group of **generalized interval exchange transformations** (GIET). Using this dictionary, we can probably use without any assumption on the group the recent result of MALICET and MILTON [71] after which one can find an invariant Radon measure.

Task 18 (*). *Prove that if a group acts freely on a topological real tree, then it preserves a non-trivial Radon measure.*

Note that a main difference compared to the classical case of actions on the line is that a group action may admit different invariant Radon measures. A more challenging problem is to generalize the result of Solodov:

Theorem 3. *If G is a subgroup of $\text{Homeo}_+(\mathbb{R})$ whose action has no global fixed point, and any of its non-trivial elements has at most one fixed point, then G quasi-preserved a non-trivial Radon measure μ on $\mathbb{R}$ (that is, any element of G takes μ to a multiple of μ).*

It is natural to extend Solodov's result to the setting of actions on topological real trees with at most one fixed point. This kind of action appears for instance in the work of Calegari and Loukidou [22]. It could be that the exact same statement holds, but perhaps there may be some issue coming from the fact that the action is not irreducible (and a technical problem is that there are several nuances of irreducibility/minimality when discussing group actions on real trees, see for instance [50]). At the moment, a safe statement would be the following:

Task 19 ()**. *Assume G acts on a topological real tree with at most one fixed point, then the action is semi-conjugate to an affine action on a Λ -tree.*

As for the previous task, the invariant affine structure may not be unique. Recall that a Λ -tree is a topological real tree X equipped with a Λ -valued distance function $d: X \times X \rightarrow \Lambda$, where $(\Lambda, <)$ is an ordered abelian group (the order on Λ allows one to express the triangle inequality for d). The point is that a Λ -tree admits way more affine actions, since the group of order-preserving automorphisms of $(\Lambda, <)$ is in general larger than that of $(\mathbb{R}, <)$. Let us mention here the foundational work of LIOUSSE [68].

A concluding line of research has been suggested by Navas: the classical correspondence between left-orderable groups and groups of homeomorphisms of the line has been generalized to groups of homeomorphisms of real trees by Horak and Stein [53], in terms of *left-invariant partial orders*.

Task 20 (*). *Establish a dictionary between properties of the left-invariant partial orders of Horak and Stein and dynamical properties of group actions on topological real tree.*

I.A.3. RT3: New tools for groups acting on surfaces

Objective 6: Classification of isometries of the fine curve graph Fix a closed orientable surface S which is not the sphere. The **fine curve graph** of S is a graph which is defined as follows. The vertices of this graph are the essential (i.e. which do not bound a disk) simple closed curves of S . Put an edge between two vertices if the corresponding curves are disjoint. By setting that each edge has length one, we obtain a metric space. Observe that the group of homeomorphisms of S acts faithfully by isometries on this graph. Indeed, if two curves are disjoint, their images by a homeomorphism are still disjoint. The definition of the fine curve graph in the case of the 2-torus $\mathbb{T}^2$ needs to be adapted in order to obtain a connected graph: in this case, we put an edge if the intersection of the two curves consists of at most one point.

This graph was introduced recently by Bowden, Hensel, and Webb in [10]. Their definition was inspired by the definition of the classical curve graph, which has a similar definition but with isotopy classes of curves as vertices instead of actual curves. Similarly to the classical case, the authors prove that the fine curve graph of S is *Gromov-hyperbolic*, which is an important property in geometric group theory (it is a property involving

only *metric* properties of the space). The study of the classical curve graph had a considerable impact on the study of the mapping class groups, the groups of connected components of the space of homeomorphisms of surfaces. Hence we can expect this graph to have similar *impact on the study* of the identity component $\text{Homeo}_0(S)$ of the group of homeomorphisms of S . Using this Gromov-hyperbolicity and techniques by Bestvina and Fujiwara, Bowden, Hensel, and Webb deduce that the *commutator length* (the number of commutators needed to write a given element as a product of commutators) *is unbounded* on $\text{Homeo}_0(S)$, thus answering a question asked by Burago, Ivanov and Polterovich 15 years ago. The definition of the fine curve graph led to many other works, indicating the way for a systematic study of groups of surface homeomorphisms. The first fundamental step is the following.

Goal 1. *Understand, or ideally establish a **dictionary**, between the dynamics on S of a homeomorphism of S and the type of isometry of its action on the fine curve graph.*

Recall that isometries of a metric space can be classified into three types, with names referring to isometries of the hyperbolic plane. An isometry ϕ of a metric space (X, d) is

- *loxodromic* (or *hyperbolic*) if there exists a point x of X such that $\lim_{n \rightarrow +\infty} \frac{1}{n} d(\phi^n(x), x) > 0$ (this limit always exists, and does not depend on the choice of $x \in X$);
- *elliptic* if ϕ has a bounded orbit (equivalently, all orbits are bounded);
- *parabolic* otherwise. In this case, all ϕ -orbits go to infinity at a sublinear speed.

This classification has important consequences on the action of an isometry on the **boundary at infinity** ∂X of a Gromov-hyperbolic space X : a loxodromic isometry has exactly two fixed points in ∂X , one attractive and one repulsive, and a parabolic fixed point has exactly one fixed point in ∂X .

A first connection in this direction of Goal 1 has been described in [9], by Bowden, Hensel, Mann, MILTON, and Webb:

Theorem 4. *An isotopically trivial homeomorphism of the 2-torus acts loxodromically on the fine curve graph if and only if its rotation set has nonempty interior.*

The **rotation set** is a compact convex subset of $\mathbb{R}^2$ which encodes the asymptotic rotation speeds around the torus: if $\tilde{f}$ is a lift of a torus homeomorphism to $\mathbb{R}^2$, then $\text{rot}(\tilde{f})$ is the accumulation set of vectors of the form $\frac{1}{n}(\tilde{f}^n(x) - x)$, for $x \in \mathbb{R}^2$ and $n \rightarrow +\infty$. The authors also prove that this condition is equivalent to saying that the homeomorphism is *isotopic to a pseudo-Anosov homeomorphism relative to a finite invariant subset* of the surface. The notion of pseudo-Anosov diffeomorphisms is very important to study the isotopy classes of homeomorphisms but will not be defined here. In the article [47], GUIHÉNEUF and MILTON generalized this result for higher genus surfaces. This generalization required to study some properties of the *ergodic rotation set* associated to a surface homeomorphism (see Objective 10), which plays the same role as the classical rotation set in the case of the torus. These two works give a **complete dynamical description** of the homeomorphisms isotopic to the identity acting **loxodromically** on the fine curve graph. In the case where the surface is the torus, a similar characterization is also made in [9] in the case where the homeomorphism is not isotopic to the identity. However, the following question remains unsolved.

Task 21 ().** *Determine which reducible homeomorphisms act loxodromically on the fine curve graph.*

Here a homeomorphism is called *reducible* if it is isotopic to a homeomorphism which preserves a finite set of simple closed essential curves but it is not isotopic to a periodic homeomorphism (i.e. a finite order homeomorphism). It is expected that they should be isotopic to a pseudo-Anosov homeomorphism relative to a finite subset. This question is related to a deeper study of rotational properties of such homeomorphisms (see Task 32).

In [48], GUIHÉNEUF and MILTON characterize the homeomorphisms of the 2-torus which act **elliptically** on the fine curve graph of the torus, as those for which all orbits stay at a bounded distance from a rational rotational direction. To prove it, they use a *new rotational object* which is adapted to the situation and allows to see orbits with a sublinear speed. The elliptic case of higher genus surfaces has not been solved yet.

Task 22 ().** *Characterize dynamically the homeomorphisms of higher genus surfaces that act elliptically on the fine curve graph.*

It is expected that such homeomorphisms have an invariant subset which looks like a simple closed curve, which should be equivalent to having so-called *bounded deviations* with respect to some closed simple geodesic. The most natural approach to this question is to generalize the work done for the torus, adapting the definition of homotopic rotation set of [49] to detect sublinear behaviours.

Objective 7: Boundary of the fine curve graph and the metric WPD property For a given Gromov-hyperbolic metric space, one can consider its **boundary at infinity**. In analogy to what has been done for the classical curve graph, it will be very helpful to represent geometrically boundary points:

Task 23 ().** *Represent boundary points of the fine curve graph as geometric objects defined on the surface.*

More precisely, we expect all the points on the boundary of the fine curve graph to be represented by an **invariant partition** of the surface. For this, we want to interpret each curve as a functional on the space of all curves (a curve α is identified with the functional which, to a curve β , associates the number of essential connected components of $\beta - \alpha$, that is the connected components which join to different lifts of α in the universal cover), in the same spirit as Thurston's compactification of the Teichmüller space.

There are some partial results in this direction. In [11], Bowden, Hensel, and Webb were able to describe some boundary points in terms of *foliations* and *geodesic laminations*. More recently, in the case of the torus, Hensel and LE Roux have related boundary points to *rotation sets* [52]: they prove that, if two homeomorphisms isotopic to the identity have the same stable (or unstable) fixed point on the boundary at infinity, then their rotation sets are similar.

To describe all the points that are fixed by homeomorphisms acting loxodromically on this graph, MILTON wrote recently a new proof of an old result by Fathi [72]. This result states that to any homeomorphism isotopic to a pseudo-Anosov homeomorphism are associated two invariant partitions of the surface, one stable and one unstable, which are in bijection with the leaves of the stable and the unstable foliations of the pseudo-Anosov homeomorphism, and such that the action of the homeomorphism on the (un)stable partition is the same as the action of the pseudo-Anosov on the (un)stable foliation. These partitions should represent fixed points on the boundary at infinity of a homeomorphism isotopic to a pseudo-Anosov homeomorphism. However, Fathi's construction is not explicit enough to prove it while, with the current technology, it should be possible to prove it with Milton's construction (a work in progress by Hensel, LE Roux and Mann should allow to do it).

A concrete way to tackle Task 23 is to understand the examples produced by EINABADI [30] of parabolic homeomorphisms of the torus which do not preserve any kind of generalized foliation and do not act properly on the fine curve graph. However, these examples could have a more degenerate invariant partition, which would represent the fixed point at infinity of this element.

In [52], Hensel and LE Roux proved a **metric WPD property** for the homeomorphisms of the torus isotopic to the identity which act loxodromically on the fine curve graph. This property extends the classical WPD (weak proper discontinuity) property for group actions on Gromov-hyperbolic spaces. The only difference is that the finite subsets which appear in the classical definition are replaced by bounded subsets. This property is the main tool for the whole article, and they deduce the aforementioned link between the rotation set and the boundary at infinity from this property. They also obtain a result about the **stable commutator length** of homeomorphisms with the help of this property. Recall that for an element g in a perfect group G , the stable commutator length of g is defined as the limit as $n \rightarrow +\infty$ of the commutator length of g^n divided by n . Positivity of the stable commutator length of an element of a perfect group G is equivalent to the existence of a homogeneous quasi-morphism $q: G \rightarrow \mathbb{R}$ such that $q(g) \neq 0$. Using the metric WPD property and famous techniques by Bestvina and Fujiwara, Hensel and LE Roux proved that any homeomorphism of the torus isotopic to the identity whose rotation set has nonempty interior and no point symmetry, has *positive stable commutator length*. In the same vein, Choi obtains in [23] a variant of metric WPD property for the action of pseudo-Anosov elements to construct new quasi-morphisms on the group of homeomorphisms of surfaces isotopic to the identity, namely unbounded quasi-morphisms which do not vanish on elements whose fixed point set is a disk. Those works lead to the following natural tasks, which are related to each other.

Task 24 ().** *Extend the results on the metric WPD property to higher-genus surfaces.*

Task 25 ().** *Describe all surface homeomorphisms whose stable commutator length is positive.*

Objective 8: Variants of fine curve graphs Here we describe two projects that are well suited for one of the post-docs we plan to hire. First, it would be natural to define fine curve graphs in the case of surfaces *with marked points and boundary components*. This should include not only nonperipheral simple loops but also arcs joining marked points or boundary components.

Task 26 (*). *Define a relevant analog of the fine curve graph for surfaces with marked points or boundary components.*

This should be a relatively easy task, as it should consist combining the ideas of the fine curve graph theory with the ones of the classical curve graph. The second project is about the **projection complex** of Bestvina, Bromberg, and Fujiwara [3,5], built from the curve graphs of all the sub-surfaces of S in the classical setting. It is a more powerful object, which, for instance, gives finer criteria for having *positive stable commutator length* [4] or for two elements to *generate a free group*.

Task 27 ()**. *Define a relevant fine projection complex and provide applications.*

Objective 9: First structure results for group actions on surfaces It is reasonable to expect that the fine curve graph and what we understand from its boundary at infinity will be helpful to deduce properties about group actions on surfaces. Margulis proved a **weak Tits alternative** for groups of homeomorphisms of the circle (see also Ghys [39]): such a group preserves a Borel probability measure on the circle or contains a copy of F_2 . Since then, a very challenging problem is to extend this for groups of surface homeomorphisms: the fine curve graph seems the appropriate tool to solve it.

Task 28 ()**. *Use the description of the boundary of the fine curve graph to provide a weak Tits alternative for groups of homeomorphisms of surfaces.*

There are already partial answers to this question. In [11], the authors use the fact that foliations of pseudo-Anosov homeomorphisms are points at infinity of the fine curve graph to deduce that if a group of homeomorphisms of a surface *contains a pseudo-Anosov homeomorphism*, then it contains a copy of F_2 or it preserves one of the foliations of the pseudo-Anosov homeomorphism. Moreover, in [52], the authors prove that a group of homeomorphisms of the torus isotopic to the identity which *contains a homeomorphism whose rotation set has nonempty interior* either contains a free nonabelian subgroup or preserves either the stable partition or the unstable partition of f . First, it is expected that the result in [72] which gives invariant partitions for homeomorphisms isotopic to (relative) pseudo-Anosov homeomorphisms, and the identification of these partitions with boundary points of the fine curve graph should allow us to generalize the result in [10] to the case of groups which contain at least one element which is isotopic to a relative pseudo-Anosov homeomorphism.

Another aspect relating group actions on surfaces and the fine curve graph is the study of the **commutation relation**. The first natural task is the following.

Task 29 ()**. *Describe centralizers of surface homeomorphisms.*

For surface homeomorphisms which act loxodromically on the fine curve graph, an element of its centralizer preserves the two fixed points on the boundary at infinity of the hyperbolic element, which should be represented by the partitions given by [72]. Hence we expect the centralizer to be essentially trivial, meaning that it should be virtually the group generated by the loxodromic element, up to a subgroup of elements supported in a common disk. For the centralizers of element which act parabolically on the fine curve graph, one has to understand first the boundary points which are given by such elements. If, as expected, those boundary points are given by partitions of the surface, then we can use those partitions to understand the centralizer of our element. In the case where the homeomorphism acts elliptically on the fine curve graph, our homeomorphism should preserve an essential compact invariant subset, which will allow us to describe the centralizer of such an element. Naturally, understanding the commutation relation will then allow to attack the following task.

Task 30 (*)**. *Study actions of nilpotent groups or solvable groups on surfaces.*

More generally, when a group action has a fixed point on the boundary of the fine curve graph, one may expect in many cases to have an induced action on some one-dimensional object (although not necessarily a nice one-manifold). In such situations, the study proposed in Objective 5 may help to get structure results for these actions.

Objective 10: Dynamics of homeomorphisms of higher genus surfaces When discussing the previous problems, we have seen that the **rotation theory** of surface homeomorphisms plays a key role. The definition of rotation set introduced in Objective 6 can be generalized to surfaces S of genus $g \geq 2$: it is a connected and compact subset of the homology group $H_1(S, \mathbb{R}) \simeq \mathbb{R}^{2g}$. The *ergodic rotation set* is a subset of the rotation set formed by the rotation vectors of points that are typical for some ergodic measure. In [41], García, GUIHÉNEUF and Lessa prove a structure theorem for ergodic rotation sets of homeomorphisms of higher genus surfaces that are homotopic to the identity. This work extends to higher genus numerous results that were only known for the torus, for example a criterion of positivity of topological entropy, or a result of realization of rational points in the interior of the rotation set by periodic orbits. This classification opens two main research directions.

Task 31 ().** *Classify (not ergodic!) rotation sets of homeomorphisms of higher-genus surfaces that are homotopic to the identity.*

It is expected is that rotation sets should be unions of at most $m(g) \in \mathbb{N}$ convex sets, where $m(g)$ is a constant depending only on the genus g of the surface, growing exponentially wrt g . If this holds, then one could seek for the optimal constant $m(g)$. This conjecture has been verified by GUIHÉNEUF in [46] for homeomorphisms whose rotation set spans the whole homology as a vector space, and by Alonso, Brum, and Passeggi on an open and dense subset of the group of homeomorphisms. In [46], GUIHÉNEUF also shows that there are rotation sets of homeomorphisms that are not the rotation set of any area-preserving homeomorphism. This suggests to attack Task 31 **focusing on some classes of dynamics**, like area-preserving maps, hamiltonian maps (this case was solved for the torus by Le Calvez and Tal), time-1 maps of flows (the case of the torus has been described in a work of Franks and Misiurewicz), etc.

Task 32 ().** *Define and classify transverse ergodic rotation sets of reducible homeomorphisms of higher genus surfaces.*

If a homeomorphism is homotopic to a finite product of Dehn twists along pairwise disjoint simple closed curves, it is possible to define a *transverse rotation set*. This task would consist in defining a good ergodic version of it (probably easy) and try to classify them, as in the classification of [41]. This is the natural way to attack Task 21. The technique we plan on using for tackling these last two tasks is the **forcing theory** of Le Calvez and Tal [64], as it was done in e.g. [41, 46]. The principle of [45] consisting in importing ideas from hyperbolic dynamics to this particular context should generalize to the more general context of homeomorphisms whose rotation decomposition of [41] is not only made of “hyperbolic classes”. A first step towards Task 32 is performed in a work in progress by GUIHÉNEUF, Le Calvez, Passeggi and Tal, which classifies nonwandering surface homeomorphisms of zero topological entropy.

I.B. Position of the project as it relates to the state of the art

In this project we describe several problems about group actions on manifolds of dimension one and two, inspired by the several results that have been produced in the last 10 years, and which suggest **several new directions for the field**. The members of the project have contributed *actively* in the development of the field, and the problems described here are partially inspired by these contributions.

In summary, several problems described in **RT1** are classical, and *very challenging*; we hope to get new insights with the help of the most recent tools, many of which have been developed by members of the project. The situation is different for **RT3**: there is a *very clear plan* on what is needed for translating properties of the action on the fine curve graph into properties for the action on the surface; this requires a *deep knowledge* on topological dynamics on surfaces, especially *rotation theory*, and the members of the project that will work on this task are among the best international experts. Finally, **RT2** is our most *innovative* and *exploratory* research track: this is the one where we expect strong interactions between all members of this proposal, especially for Objective 5.

I.C. Methodology and risk management

Methodology and scientific agenda The project is expected to last from Oct. 1, 2026 to Sep. 31, 2031. The members of the project have consolidated experience in working together, with regular scientific collaborations

and tradition of organizing scientific events. Strong of this experience, the project will advance with regular meetings, visits, and invitations of international experts.

Besides working on the tasks described before, we will do our best to disseminate the results and advertise the topic of dynamical group theory to the young generation of researchers. This will hopefully bring new ideas and open unexpected directions. For this, when organizing events, we will encourage the participation of young participants, with financial help and carefully-chosen scientific program. We will also write *surveys* and *monographs* when appropriate: for instance, TRIESTINO is writing a survey paper on smoothability and non-smoothability of one-dimensional actions for *Journal of Modern Dynamics*; GUIHÉNEUF has started writing a monograph on the rotation theory of surface homeomorphisms, based on his habilitation thesis, and he has a plan to write with LE ROUX an introduction to the fine curve graphs based on the material they prepared for a master course in Sorbonne Université.

In the course of the 5 years of the project, we will have regular exchanges with *international experts*. Dynamical Group Theory is particularly developed in the USA, South Korea, and countries of South America. The members of the project collaborate with the best specialists of the field. Here is a partial list: Andrés NAVAS, Cristobál RIVAS (Chile); Juan ALONSO, Sébastien ALVAREZ, Joaquín BRUM, Nancy GUELMAN, Alejandro PASSEGGI (Uruguay); Thomas KOBERDA, Yash LODHA (US); Hyungryul BAIK, KyeongRo KIM, Sang-hyun KIM (South Korea); Sebastian HENSEL (Germany). Note that Kathryn MANN and Andrés NAVAS, two leading figures of the field and very close collaborators of the members of this project, will be working in France starting from September 2026; they will certainly take an active part to the success of this project, but they are not formally declared as members because they will (hopefully) have access to different source of funding: MANN, who has recently joined the IMJ-PRG as a CNRS researcher, has applied for an ERC grant; NAVAS will be staying in Rennes for 3 years, funded by the ANR program *Choose France for Science*.

We will have the chance of organizing the main events in centers of high international visibility, such as a **thematic month** at CIRM in Marseille, and a **Summer School in Jussieu** in Paris. We plan to have both of them in 2029, when both post-docs will be hired for the project. We have also considered organizing a special semester at CIRM, applying for a **Jean Morlet Chair**: for this, we have an agreement in principle from Sebastian HENSEL and Yash LODHA, who are natural candidates for this prestigious chair. The administrative work for such a semester can be heavy, and for this reason we prefer to organize smaller events and have more time to focus on research. Anyhow, if we will estimate that there are good conditions for the organization of a Jean Morlet Chair, we will not hesitate in applying for it and find possible co-funding.

In order to share the advances on the project and share ideas on possible new directions, we will have regular meetings for all the members of the project. A good format for this are 3-day meetings every year, to which we will also invite interested young researchers.

For a summary of the agenda, we refer to the Gantt chart (Table 1).

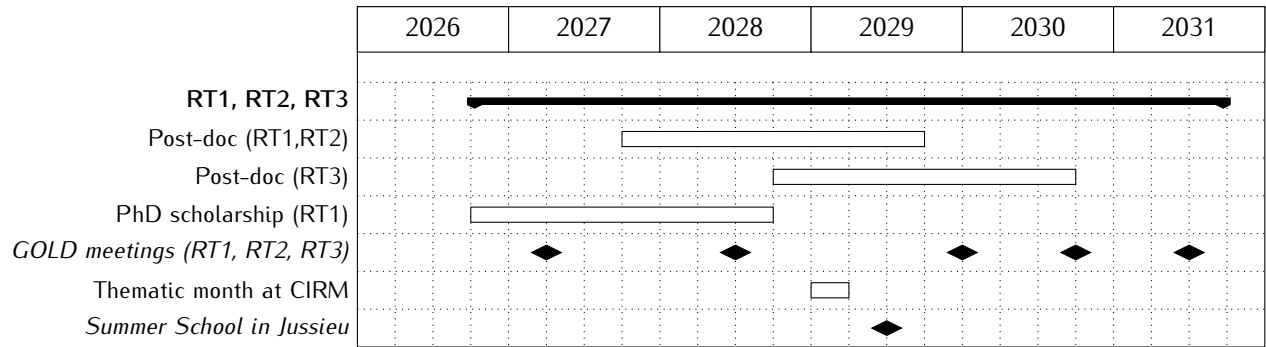


Table 1: Gantt chart

Risk management The main risk is that we could not be able to make satisfactory progresses at the desired speed. For this reason, we are presenting a long list of tasks, of different level of difficulty: well-known conjectures, new directions suggested by recent results, or consolidating a new growing theory. It is also highly likely that unexpected results will be obtained while working on our objectives, and we will encourage the members to explore also new promising directions. The regular meetings will be perfect occasions for

mutual help and for transforming partial ideas and advances into definite results: the consortium is formed by researchers with solid scientific production, with a strong tradition of collaboration.

II. Organisation and implementation of the project

II.A. Scientific coordinator and consortium

Michele TRIESTINO, professor at Aix-Marseille Université since 2026 and junior fellow of the Institute Universitaire de France, will be the **coordinator** of the project. He is one of the leading figures in the study of group actions on one-manifolds and orderable groups; in his works he connects ideas from dynamical systems, topology and geometric group theory. He has a strong experience in leading research projects involving several people: he was the PI of the ANR project **Gromeov** (2019–2025), French coordinator of the MATH-AmSud project *Dynamical Group Theory* (2022–2023), and currently PI of the MATH-AmSud project *Interactions between groups and dynamics* (2026–2027). He has also a strong experience in organizing scientific events, of different nature: for instance, he has recently joined the organizing committee for the series of conference **Beyond Uniform Hyperbolicity**, with the last event in 2025 at ICTP having nearly 200 participants over two weeks. He has co-supervised the PhD thesis of 2 students, and he is currently (co-)supervising 3 students on the topics of this proposal.

As coordinator of the project, he will be charged of setting out the scientific agenda and logistics of meetings and events. He will also work on the international visibility of the project, encouraging collaborations with international researchers, and advertising the events to young researchers.

Partner 1 (Aix-Marseille Université)

1. *Partner's scientific leader*: Michele TRIESTINO (PR, Marseille & IUF fellow). He will work on RT1 and RT2, and he will be also the **scientific coordinator for RT1**.
2. Isabelle LIOUSSE (MCF, Lille) is a specialist on group actions on low-dimensional manifolds, under different point of views. Considering her unique expertise in groups of (generalized) interval exchange transformations [42–44], her contribution will be highly useful for the tasks described in Objective 5.
3. Nicolás MATTE BON (CR CNRS, Lyon 1) has several important works on group actions on one-manifolds [15–17, 61, 75], and his expertise in group theory goes far beyond this somehow restricted topic. He will work on RT1 and RT2, and he will be the **scientific coordinator for RT2**.
4. Maxime WOLFF (PR, Toulouse) works on a broad range of topics in topology and geometry, from quantum topology to Lorentzian geometry. He has important publications on low-dimensional group actions, highly related to this project [65, 66, 73, 74]. His expertise will be important for all research tracks.

Partner 2 (IMJ-PRG)

1. *Partner's scientific leader*: Pierre-Antoine GUIHÉNEUF (MCF, Sorbonne Univ. & ENS). He is a recognized expert in topological surface dynamics and in particular rotation theory and forcing theory. The series of works [41, 45–49] are fundamental contributions to the long-term projects described as Objectives 6 and 10. He will work on RT3.
2. Hélène EYNARD-BONTEMPS (PR, Sorbonne) is a specialist of smooth one-dimensional dynamics, in connection with codimension-one foliations [8, 31, 34], and distortion properties [32, 33]. Her expertise will be particularly important for Objective 3 of RT1 and Objective 4 of RT2.
3. Frédéric LE ROUX (PR, Sorbonne) is an expert in topological surface dynamics. In relation with the project, he has applied techniques from topological dynamics to the study of group actions on surfaces. He has several recent works in relation with RT3 of this proposal [35, 52, 65, 66]. He is interested in every objective of the project where his expertise can help, and especially those described in RT3.
4. Dominique MALICET (MCF, Marne-la-Vallée) is an expert in dynamical systems and probability, with contributions to the field of one-dimensional group actions highly relevant to Objective 5 of the project [70, 71].
5. Emmanuel MILTON (MCF, Nice) is an expert of topological dynamics on surfaces, and he studies groups of homeomorphisms of manifolds by topological methods and with techniques coming from geometric group theory. His recent works [9, 47–49, 72] are fundamental for the objectives described in RT3. In

addition, his works [70,71] are related to Objective 5 of RT2. He will be the **scientific coordinator for RT3**.

PhD students of members of the project will naturally take part to the activities: Pedro ALVES (supervised by GUIHÉNEUF and MILITON), Guillaume DE CARPENTIER (supervised by TRIESTINO, starting in September 2026), Martín GILABERT VIO (supervised by MATTE BON; starting from September 2026 he will be post-doc in Rennes), Sandro PUPULIN (supervised by Luisa PAOLUZZI and TRIESTINO), Théo VIROT (doctorant TRIESTINO).

Implication of the scientific coordinator and partner's scientific leader in on-going project(s)

Name of researcher	p.month	Call, funding agency, grant allocated	Project's title	Scientific coordinators	Start-end
TRIENTINO	ND	Math AMSUD (CNRS, MEAE, ANID, ANII), 31k€	IBGD - Interactions between groups and dynamics	M. Triestino, N. Matte Bon, J. Brum, C. Rivas	2026-27
TRIENTINO	15p.month	ANR PRC 348k€	AnoDyn ANR-24-CE40-5065-01	P. Dehornoy	2024-29

II.B. Implemented and requested resources to reach the objectives

As for most of mathematical projects, scientific exchanges represent the principal mean of achievement. The scientific agenda has been described in §I.C, in relation with the objectives of the proposal. Whenever possible, we will ask the co-participation of exterior partners, such as the réseaux thématiques of CNRS **Tétraèdre de Platon** and **Topologie Algébrique et Géométrie** for national events, or the local funding in Marseille available from **CIRM** and **FRUMAM**. The grant attached to the IUF fellowship of TRIESTINO can be helpful for co-funding events (other members have applied for IUF fellowships this year, and this may lead to other sources of co-funding). Many of our international collaborators (such as S.-h. KIM, K. KIM, NAVAS, RIVAS, LODHA, HENSEL – see the list in §I.C) have recurrent grants which can be used for visits, or co-funding events. We can also rely on institutions for specific international cooperations, such as **IRL IFUMI** for Uruguay, or **MATH-AmSud** for cooperation with South America (Uruguay and Chile have a long tradition of forming many master students with strong background on dynamics of group actions). In this direction, EYNARD-BONTEMPS and NAVAS are PI of a grant from the program **ECOS Sud Chile** (ending in 2026), while TRIESTINO is the PI of a grant from MATH-AmSud (for 2026–2027).

The requested funds amount to **447k€**, detailed as follows (read Marseille=*Partner 1 (Institut de Mathématiques de Marseille - UMR CNRS 7373)*, IMJ=*Partner 2 (Institut de Mathématiques de Jussieu-Paris Rive Gauche - UMR CNRS 7586)*).

Staff expenses A 2-year *PhD scholarship* (**83k€** Marseille) for the current first-year PhD student Sandro PUPULIN: the project of his thesis is related to Objective 2, but unfortunately he has started his thesis without a scholarship.

We also ask a *2-year post-doc* for RT1 and RT2 (**117k€** Marseille) and a *1-year post-doc* for RT3 (**88,28k€** IMJ). Not only these scholarships will be important for achieving some of the objectives, but they will help attracting young researchers to the growing subject of Dynamical Group Theory. There are several young researchers that may be interested in the post-doc offers, and who are ideal candidates for working on the tasks of this project: for RT1 and RT2 we can mention Tommy Wuxing Cai, Leonardo Dinamarca, Maximilano Escayola, Martín Gilabert Vio, and KyeongRo Kim, while for RT3 a non-exclusive list is given by Nastaran Einabadi, Alejo García Sassi, Yongsheng Jia, Yusen Long, Favio Pirán, and Nelson Schuback.

General and administrative costs & other operating expenses We plan to organise several meetings and events during the 5 years of the project.

- A **thematic month** at CIRM (Marseille) on Dynamical Group Theory. This is a regular event co-organized by members of the Math department in Marseille, taking place every February, with 4 or 5 weeks of thematic schools and research conferences. We plan to have it in 2029. (**30k€** Marseille)
- A **Summer School in Jussieu** at Sorbonne Université (Paris). This is a regular event of one or two weeks organized by members of the IMJ-PRG taking place every summer around June. We plan to have it in 2029. (**10k€** IMJ)

- *3-days meetings* for members of the project and a few more young participants, with one meeting every year. We estimate a participation of around 20 people for each event (21k€ Marseille, 14k€ IMJ).

We will also need regular scientific exchanges between members of the project, or with international collaborators, besides the opportunities given by the events listed above.

- We propose to cover expenses of long-term visits, for a total of 4 months, for *invitations of international collaborators*: natural candidates are Joaquín BRUM (for RT1, RT2), KyeongRo KIM (for RT1, RT2), Alejandro PASSEGGI (for RT3), or Sebastian HENSEL (for RT3), but we will give priority to international PhD students, especially from South America, who usually have more difficult access to funding (5k€ Marseille, 5k€ IMJ).
- 20k€ for *visits* between members of the project and *participation* to other events. This will allow us to cover expenses for missions of approximately 4 weeks every year, shared between members of the project (10k€ Marseille, 10k€ IMJ).

Finally, the *administrative management* requires 13,5% of the total budget (35,98k€ Marseille, 17,18k€ IMJ).

Requested means by item of expenditure and by partner

	Partner 1 Aix-Marseille Université	Partner 2 IMJ-PRG
Staff expenses	200,55k€	88,28k€
Other costs (scientific visits and events)	66k€	39k€
Administrative management & structure costs (13,5%)	35,98k€	17,18k€
Sub-total	302,53k€	144,47k€
Requested funding		447k€

III. Impact and benefits of the project

Our project is at the interface between group theory, geometry and dynamical systems. It will impact our mathematical community in several ways, beyond the importance of the expected results:

- We will contribute to the development of the subject of Dynamical Group Theory, which has been surprisingly changing since 2020, with several important results.
- With regular meetings and travels, it will consolidate a national network of researchers in the domain, and encourage exchanges with international collaborators.
- Through this network, it will attract French and foreigner students to this active area of research. Our project features several problems that can be successfully studied by PhD students or post-docs.
- The results obtained by the members of the consortium will be disseminated as publications in mathematical journals and oral presentations. The members of the project have a tradition of privileging quality over quantity for scientific publications; whenever appropriate, we will choose open access journal. We also plan to write surveys and monographs, as we mention in §I.C, and popularization articles for *Images des Mathématiques*.
- We will maintain a *webpage* for the project, and this will increase the visibility of the obtained results and sponsored events.

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